

HYSTERESIS + BOUNDS

high: $f > \tau_{on}$ for N_{on} frames + cooldown

clear: $f < \tau_{off}$ for N_{off} frames + hold

option: T_{rec}/T_{ext} ; rejoin: T_{rejoin} ; budgets: K, R

T_{seq} spans recovery, rejoin, and emergency stops

meaningful goal progress resets K and T_{seq}

EMERGENCY STOP

hold while hazardous;
stop or bounded escape

HAZARD-CLEAR ROUTING

saved $\in \{\text{RECOVERY}, \text{REJOIN}\}$; high: $f > \tau_{on}$

high + saved $\rightarrow$ RECOVERY

high + other $\rightarrow$ PENDING

else + saved $\rightarrow$ REJOIN

else + other $\rightarrow$ NORMAL

hazard preempts nominal state logic

confirmed while still NORMAL

clear + done/progress
or option timeout

NORMAL

Nav2 tracks
the original goal

high

PENDING RECOVERY

debounce + cooldown

not high

RECOVERY

bounded masked
recovery option

high / T_{rejoin}
retry count $< R$

REJOIN

restore original goal;
verify progress

progress + low risk, or retries exhausted

NONTERMINAL PRIORITY: goal $\rightarrow$ unrecoverable $\rightarrow T_{seq}$ cap $\rightarrow$ safety $\rightarrow$ state logic

FAILED

terminal until reset

FAILURE GUARDS

unrecoverable: any nonterminal state
 T_{seq} expired: any active sequence
 K exhausted: NORMAL / PENDING

GOAL GUARD

goal reached
checked first

SUCCEEDED

terminal until reset